

CHINO, CA, USA

WMO: 722899

Lat: 33.975N

Lon: 117.636W

Elev: 650

StdP: 14.35

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 03179

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	34.3	36.6	6.3	7.8	65.3	12.2	10.5	65.9	24.5	62.7	20.6	63.3	1.7	10	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	31.7	101.1	69.6	98.2	69.0	95.1	68.0	73.6	94.2	72.1	92.1	70.7	90.1	11.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	103.0	80.7	65.2	95.5	79.2	63.7	90.4	77.1	37.4	94.7	36.0	92.3	34.8	90.0	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.8	17.2	16.0	DB	29.5	108.1	2.5	4.3	27.7	111.2	26.3	113.7	24.9	116.1	23.1	119.3
			WB	27.3	76.1	2.3	2.9	25.7	78.2	24.3	79.8	23.0	81.4	21.3	83.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.1	55.0	55.1	58.5	61.5	66.3	71.6	77.4	77.8	75.0	67.8	60.1	53.9	(6)
(7)		DBStd	10.08	5.73	5.70	5.73	5.88	5.08	5.00	4.54	4.38	5.71	5.67	5.80	5.20	(7)
(8)		HDD50	60	17	13	4	2	0	0	0	0	0	0	3	21	(8)
(9)		HDD65	1541	313	280	214	136	44	5	0	0	2	35	168	344	(9)
(10)		CDD50	5557	173	156	268	347	505	649	849	861	750	551	305	143	(10)
(11)		CDD65	1563	4	3	12	31	84	204	384	396	302	121	21	1	(11)
(12)		CDH74	19567	169	206	396	714	1184	2324	4273	4504	3446	1699	550	102	(12)
(13)		CDH80	10162	26	52	123	280	516	1145	2400	2623	1964	824	197	12	(13)
(14)	Wind	WSAvg	5.3	4.3	4.5	5.2	6.2	6.8	6.8	6.5	6.0	5.2	4.7	4.0	3.9	(14)
(15)	Precipitation	PrecAvg	11.5	2.3	2.5	1.4	1.0	0.2	0.0	0.1	0.0	0.2	0.5	1.2	1.9	(15)
(16)		PrecMax	24.4	8.4	8.4	3.4	2.9	0.9	0.1	0.9	0.2	0.5	5.5	2.2	8.5	(16)
(17)		PrecMin	4.4	0.4	0.2	0.3	0.3	0.0	0.0	0.0	0.0	0.1	0.0	0.4	0.2	(17)
(18)		PrecStd	5.4	2.2	2.1	0.8	0.6	0.2	0.0	0.2	0.0	0.1	1.2	0.5	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.6	85.7	89.5	94.4	97.3	102.1	104.1	105.2	105.7	99.0	91.4	81.6	(19)
(20)			MCWB	55.0	56.7	60.8	60.4	64.9	68.3	71.8	70.2	70.1	64.6	59.2	55.8	(20)
(21)		2%	DB	78.9	80.8	83.4	87.5	91.1	95.3	99.1	100.1	100.0	94.0	85.8	76.9	(21)
(22)			MCWB	54.6	56.0	58.8	59.9	63.9	68.2	71.0	70.1	69.6	62.5	58.1	53.8	(22)
(23)		5%	DB	74.6	74.8	78.4	82.0	85.9	90.8	95.5	97.1	96.3	89.7	80.7	72.4	(23)
(24)			MCWB	53.4	54.4	57.7	59.0	63.4	67.3	70.3	69.4	68.4	61.5	56.6	52.0	(24)
(25)		10%	DB	70.1	69.2	72.9	76.9	80.8	86.9	92.2	93.2	91.4	83.9	75.1	67.4	(25)
(26)			MCWB	51.9	53.2	56.6	58.1	62.7	66.2	69.6	68.9	67.0	60.8	55.3	51.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.9	61.1	65.0	66.1	70.7	72.3	76.6	74.9	74.7	69.7	62.9	60.2	(27)
(28)			MCDWB	71.8	72.9	81.4	85.7	89.8	93.0	94.2	97.3	97.2	87.5	76.9	69.2	(28)
(29)		2%	WB	58.2	58.7	61.8	63.0	67.4	70.1	73.9	72.8	72.6	67.2	60.8	57.9	(29)
(30)			MCDWB	68.7	71.7	77.2	80.4	86.3	90.1	93.1	94.2	94.1	84.0	76.9	66.4	(30)
(31)		5%	WB	56.4	56.9	59.7	60.8	65.2	68.4	72.2	71.3	70.6	65.1	59.3	56.1	(31)
(32)			MCDWB	66.9	69.6	73.8	77.5	82.1	87.7	90.9	92.1	90.7	81.7	73.8	64.8	(32)
(33)		10%	WB	54.6	55.1	57.9	59.1	63.0	66.9	70.6	69.9	68.7	63.3	57.8	54.2	(33)
(34)			MCDWB	64.8	65.5	69.5	73.8	78.1	84.8	88.6	90.0	87.0	77.6	70.5	62.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	26.6	25.1	25.8	26.6	25.9	28.0	30.2	31.7	31.3	29.3	29.1	26.6	(35)
(36)			MCDBR	35.8	37.2	36.2	37.6	37.1	35.9	33.9	36.2	38.4	40.2	38.5	36.2	(36)
(37)			MCWBR	16.5	16.7	16.0	15.0	14.3	12.6	11.2	11.9	12.5	14.1	15.7	17.1	(37)
(38)		5% WB	MCDWR	27.3	29.3	30.1	31.8	32.1	32.7	30.9	32.3	33.3	32.1	30.9	26.6	(38)
(39)			MCWBR	14.2	14.6	14.6	14.2	13.6	12.4	11.5	11.6	12.2	12.9	14.2	14.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.317	0.341	0.354	0.355	0.366	0.379	0.363	0.348	0.343	0.304	0.303	(40)	
(41)		taud	2.543	2.533	2.443	2.412	2.442	2.420	2.448	2.480	2.509	2.498	2.596	2.594	(41)	
(42)		Ebn at Noon	288	297	297	296	295	290	285	289	290	284	288	285	(42)	
(43)		Edh at Noon	26	29	34	37	36	37	36	34	32	30	24	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1021	1304	1753	2132	2327	2494	2418	2291	1947	1481	1137	926	(44)	
(45)		RadStd	71	128	89	99	129	131	115	59	62	93	57	65	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+2.66	+1.24	-5.89	+2.05	N/A	N/A	N/A	-414	+994	+536	(46)
(47)	Regional (51 neighbors)	+1.13	N/A	-3.06	+1.30	N/A	N/A	N/A	-202	+401	+202	(47)

Nomenclature: See separate page